

Aircraft Design 101

Evaluating and analysing the chosen concepts

CADEM0016

Unit Director: Fintan Healy (QB1.31) fintan.healy@bristol.ac.uk

Recap

Lecture 1

Turning customer needs into engineering requirements

What the aircraft must do

Lecture 2

Identifying concepts that can satisfy those requirements

*Exploring possible ways
to solve the problem*

Today

Evaluating and analysing the chosen concepts

Understanding how well it works and where it falls short.

- Customer requirements → TLARs
- TLARs define the problem
- Given the TLARs & customer requirements we can choose concepts **likely** to solve the problem
- How can we begin to analyse these concepts?

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ILOs

- Understand how to complete the initial sizing of a “*clean sheet*” aircraft configuration
- Understand how to complete constraint analysis to estimate thrust-to-weight and wing loading parameters
- Be able to apply “*top-down*” and “*bottom-up*” approaches to estimate the empty mass of an aircraft.

Initial Sizing

Clean Sheet Sizing

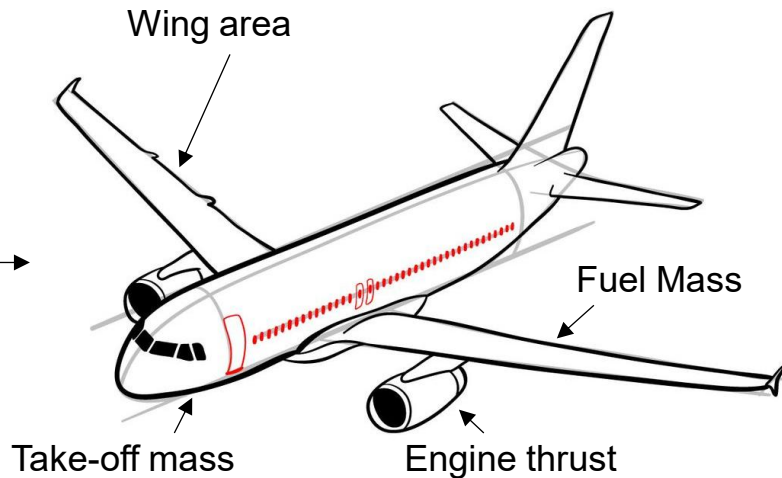
TLARs

+

Configuration

- For a given concept, we want to estimate its performance (your objective/cost functions)
 - Fuel burn
- But we are faced with a seemingly insurmountable number of unknowns
- First, acknowledge we don't “*draw the aircraft and see how far it goes.*”
- We start from the mission, and build an aircraft to:
 - fly a specific range
 - carry a specific payload

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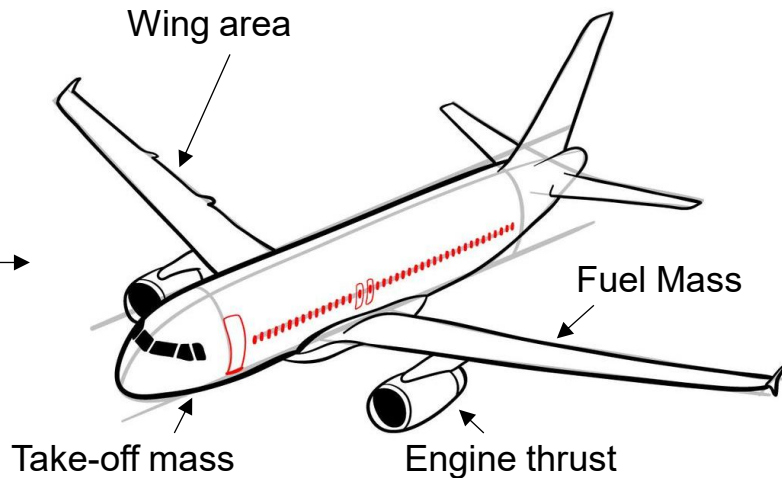
Clean Sheet Sizing

TLARs

+

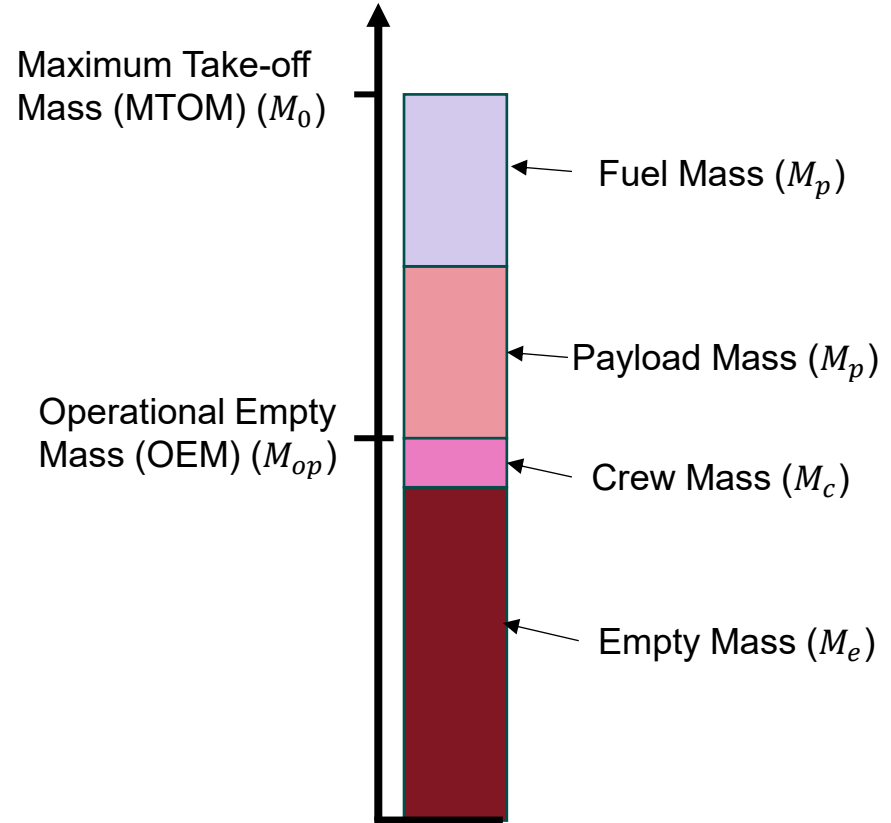
Configuration

- We're not designing an aircraft!
- We're building a tool that:
 - Matches the performance of a known design
 - We need to anchor the results in reality
 - Can be used to see how performance/cost varies as we change some higher-level values
 - We want the changes to be representative
- What would you expect if we vary these parameters?
 - Higher Aspect Ratio?
 - Fuselage width?



Mass Breakdown

- Regardless of concept. The mass of the aircraft will be made up of four components
- Payload mass and crew mass are fixed (tied to TLARs)
- We need to estimate two components:
 - The empty mass of the aircraft
 - The required fuel mass (to complete the “design mission”)



Classification of Methods

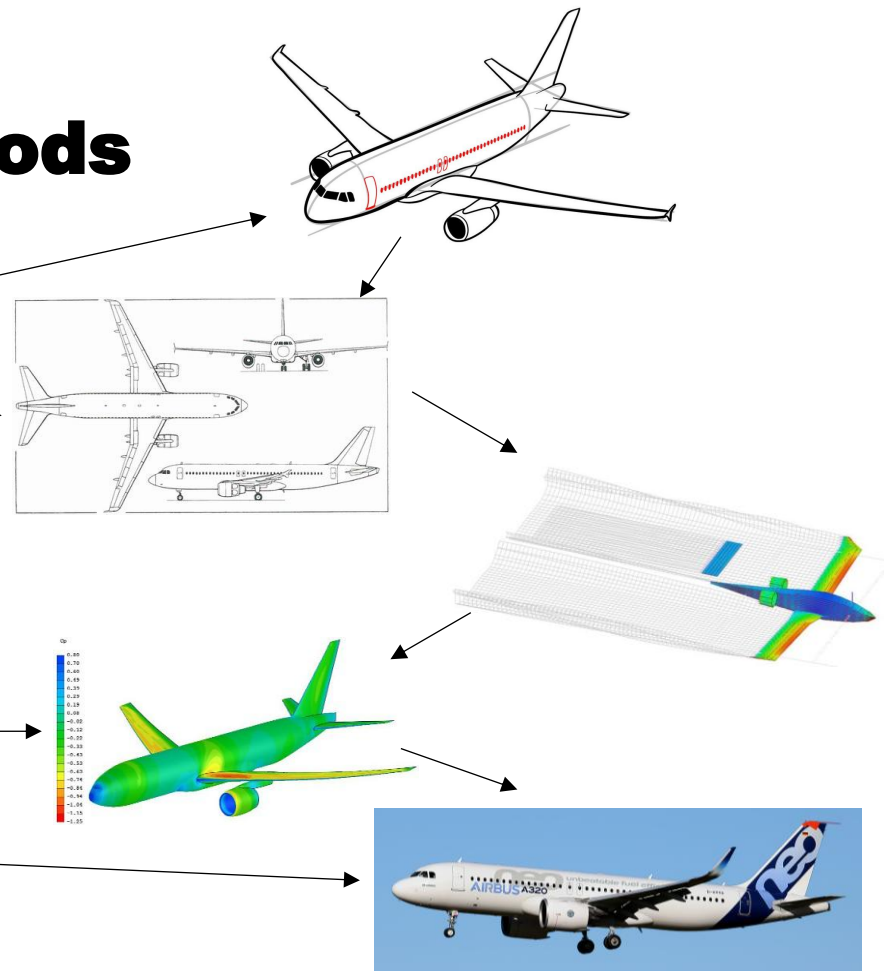
•**Class I:** Low-fidelity, geometry-free statistical weight estimation based on TLRs and historical aircraft.

•**Class II:** Medium-fidelity conceptual design using early geometry, quasi-analytical structural methods, and group-level weight statements.

•**Class II.5:** Simplified physics-based models (e.g., beam theory, vortex-lattice aerodynamics, parametric structural sizing) calibrated with empirical data to bridge conceptual and preliminary design.

•**Class III:** High-fidelity preliminary design using subsystem-level breakdowns supported by CFD, FEM, loads analysis, and digital geometry.

•**Class IV:** Fully itemised, drawing-based mass accounting of every component during detailed design.



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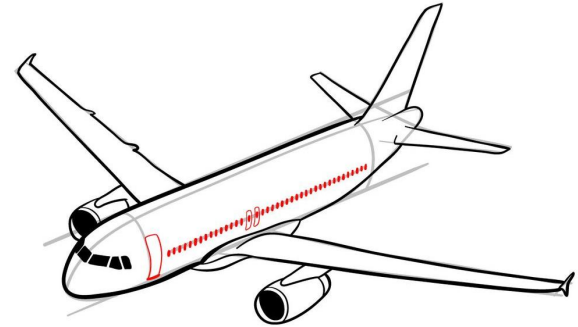
•**Class IV:** Fully itemised, drawing-based mass accounting of every component during detailed design.

▪ Top Down:

- Use simple, empirical models
- Quick “spreadsheet” calculations
- Must have good data

▪ Bottom Up

- Build up weights from components
- Use analytical tools to calculate L/D, etc
- Generally, more trusted, but more detailed and takes longer



Reducing Dimensionality

- **Sizing Philosophy**

- There are thousands of parameters you can vary on an aircraft, but we need a tractable problem!
- Three types of parameters
 - TLARs
 - Define what the aircraft should do
 - Hyperparameters
 - Define how you can vary geometry of the aircraft
 - Constants
 - Parameters you will leave constant across optimisation
- For a given set of parameters, “sizing” should be a deterministic problem
- Keep a record of your parameters
 - Don’t forget, you can promote / relegate parameters throughout the process

TLARs

Payload (Fleet level)
Range (Design Mission)
Cruise Mach Number?

....

Hyperparameters

Aspect ratio: range 8 to 20
Cruise Mach Number: range 0.5 to 0.95?
Fuselage decks: range 1 to 3?
Number of aircraft... range 1 to 100?

....

Constants

Primary material
Number of engines
C_L max?
Cruise Mach Number: range 0.5 to 0.95?

...

CLASS I SIZING

A top-down approach*

Mass Fractions

- Remember the aim: we need to estimate:
 - Empty mass
 - Fuel mass
- Normalise unknowns by the MTOM (M_0)
- We can then compare these values to historical trends

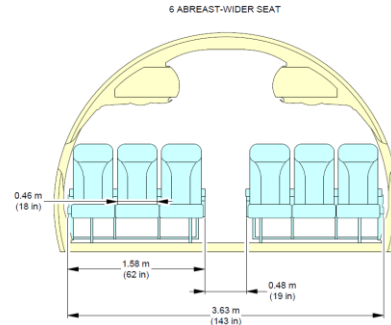
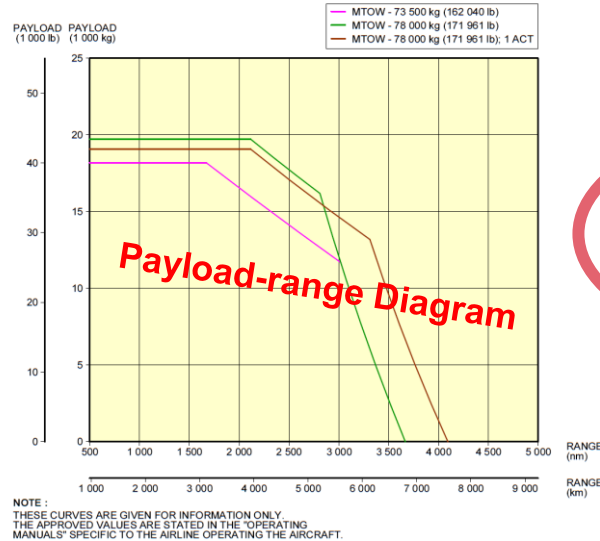
$$M_0 = M_p + M_{op} + M_f$$

$$M_0 = M_p + M_0 \frac{M_{op}}{M_0} + M_0 \frac{M_f}{M_0}$$

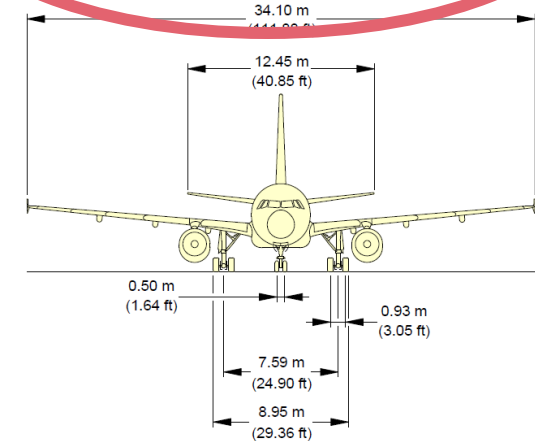
$$M_0 = \frac{M_p}{1 - \left(\frac{M_f}{M_0}\right) - \left(\frac{M_{op}}{M_0}\right)}$$

A320 “Airport Characteristics” Document

- There are many databases with historical aircraft data
- All manufacturers of large aircraft must also produce documentation

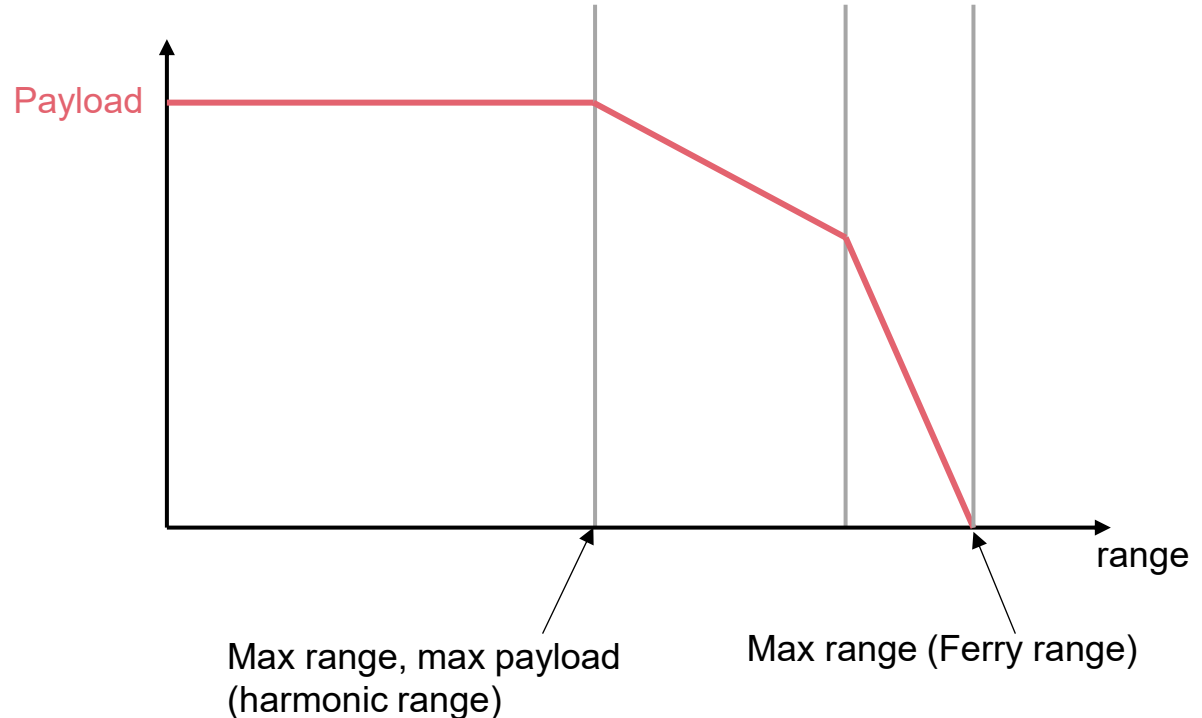


	VV000
Maximum Ramp Weight (MRW)	73 900 kg (162 822 lb)
Maximum Taxi Weight (MTW)	73 500 kg (162 040 lb)
Maximum Take-Off Weight (MTOW)	73 500 kg (162 040 lb)
Maximum Landing Weight (MLW)	64 500 kg (142 198 lb)
Maximum Zero Fuel Weight (MZFW)	60 500 kg (133 380 lb)



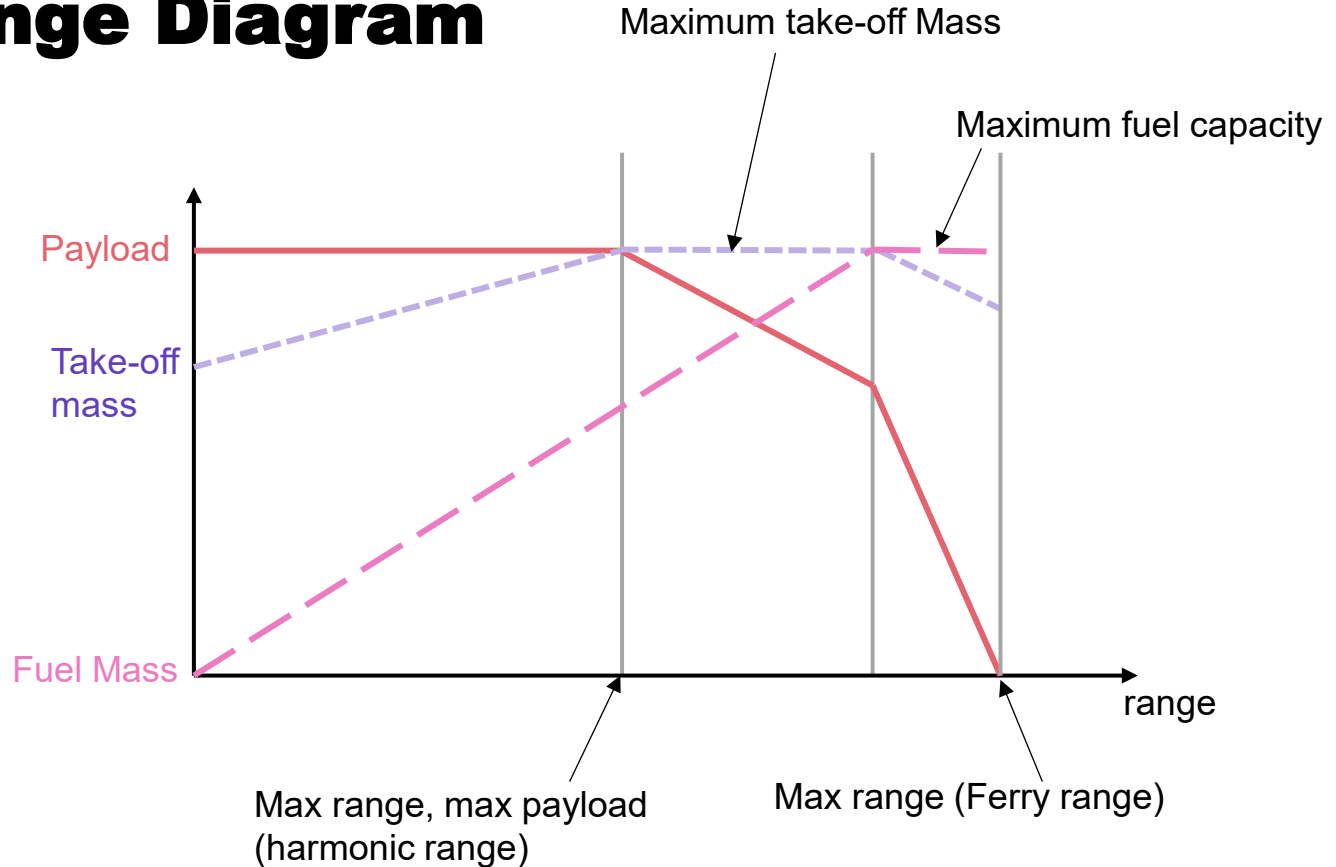
Payload Range Diagram

- The payload range diagram describes the possible range of an aircraft for different payloads
- Can extract a lot of useful information
 - Max. payload!
 - Max range



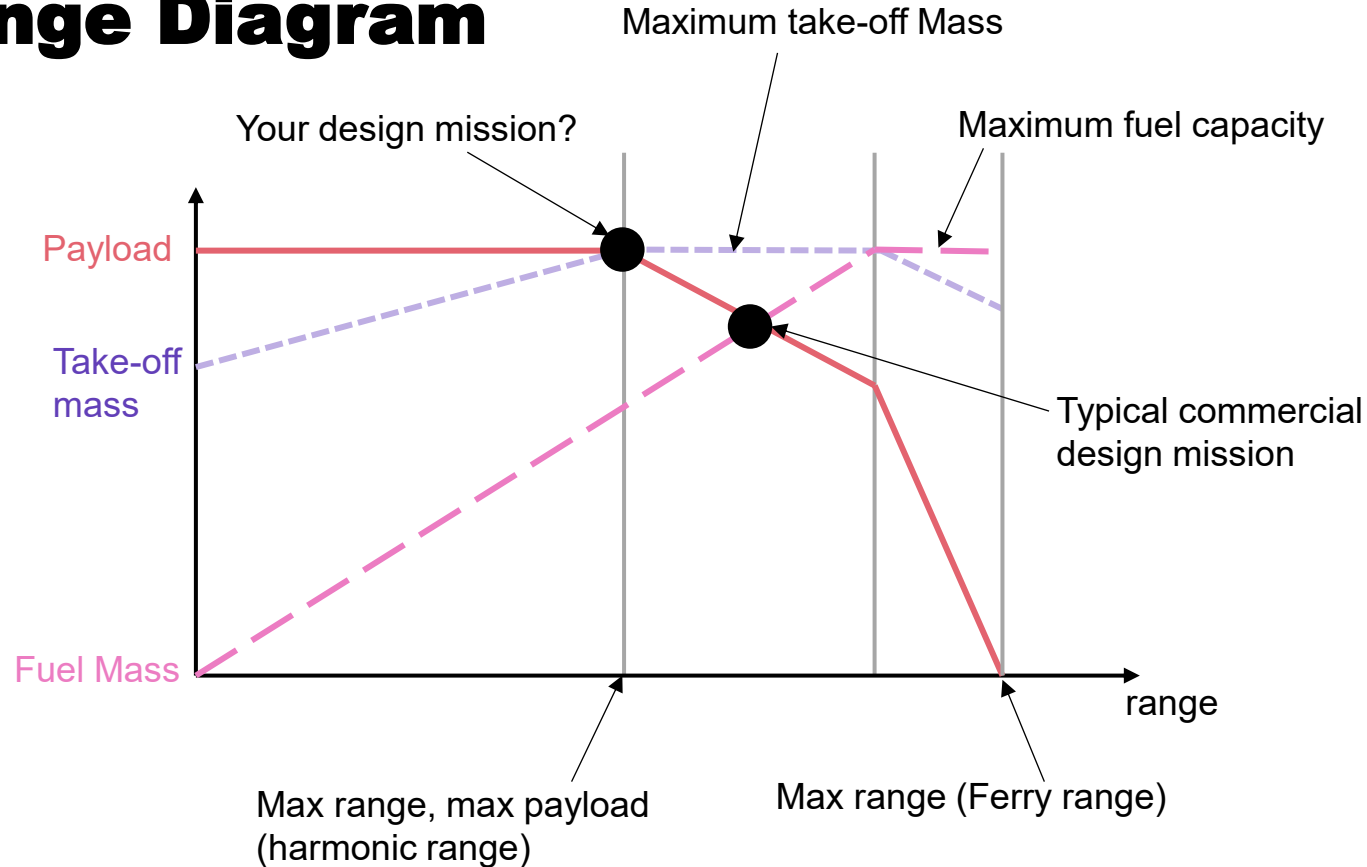
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Payload Range Diagram

- The payload range diagram describes the possible range of an aircraft for different payloads
 - Normally, only the red line
- Can extract a lot of useful information
 - Max. payload!
 - Max range
 - Fuel efficiency

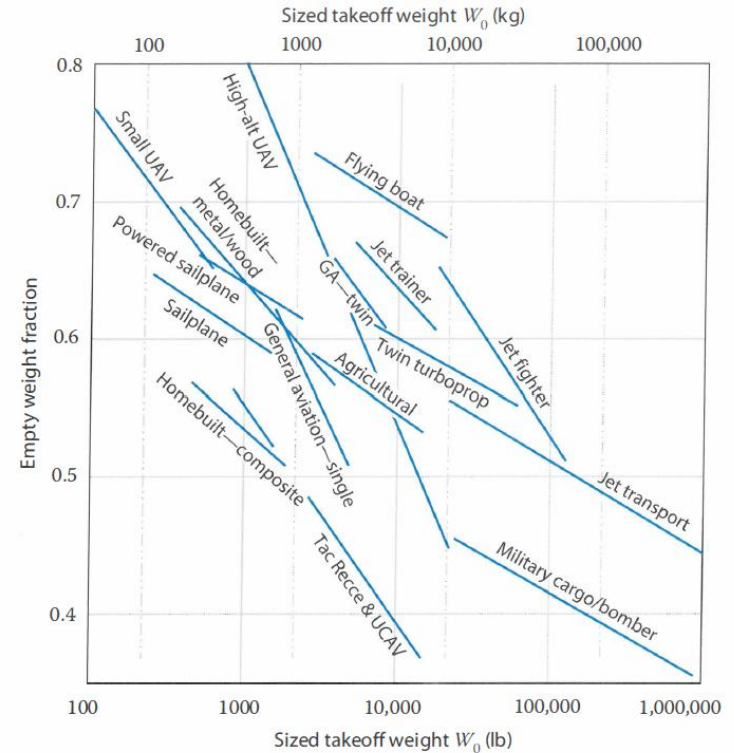


A320 Parameters

- Max. zero fuel mass – max. payload = operational empty mass
- If we assume a harmonic design mission...
 - Fuel mass $\approx$ MTOM – Maximum landing mass

What about other Aircraft?

- For **Empty mass fraction**
 - Lots of empirical relationships can be found in literature



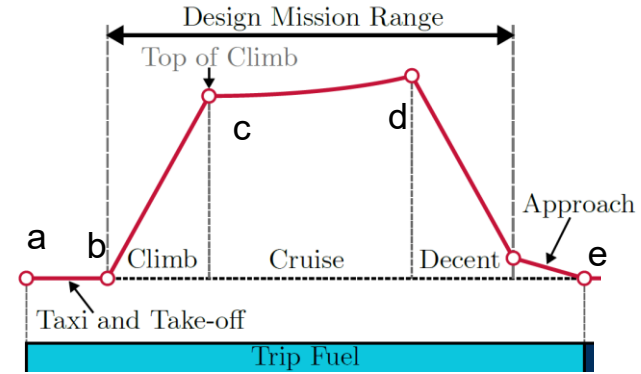
Raymer

Fuel Fraction - Mission Analysis

- To estimate fuel mass, we can estimate the fuel burn during different segments of a mission

$$M_b = M_b$$

$$M_b = M_a \frac{M_b}{M_a}$$



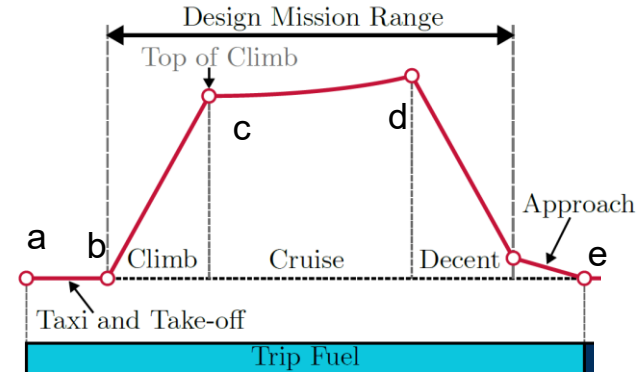
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$$M_c = M_b \frac{M_c}{M_b} = M_a \frac{M_b}{M_a} \frac{M_c}{M_b}$$



Fuel Fraction - Mission Analysis

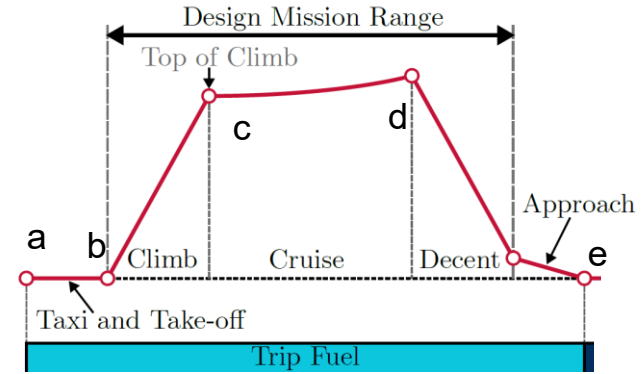
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$$M_e = M_a \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$



Fuel Fraction - Mission Analysis

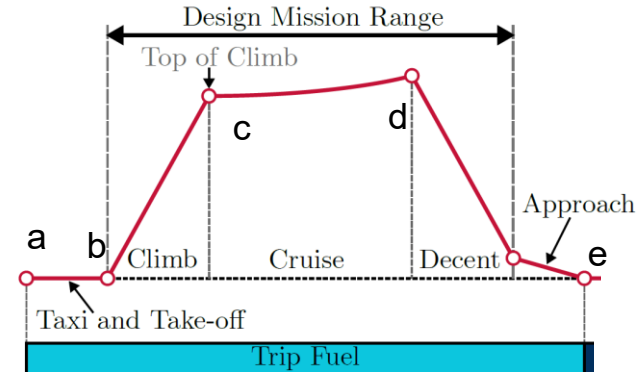
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$$M_e = M_a \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$



- The fuel fraction is therefore

$$\frac{M_f}{M_a} = \frac{M_a - M_e}{M_a} = 1 - \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$

- All we need to do now is calculate the mass fraction for each segment!

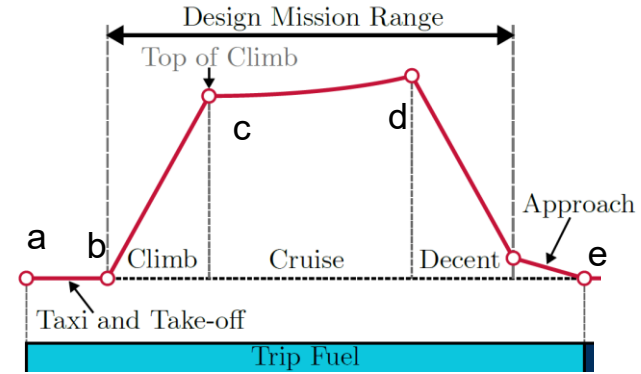
Empirical Estimates

- In lieu of better analysis, statistical values from historical aircraft can be used

Table 3.2 Historical Mission-Segment Weight Fractions

Mission Segment	(W_i/W_{i-1})
Warmup and takeoff	0.970
Climb	0.985
Landing	0.995

Raymer



$$\frac{M_f}{M_a} = \frac{M_a - M_e}{M_a} = 1 - \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$

Cruise Mass Fraction

- For cruise, we can rearrange the **Breguet Range Equation**

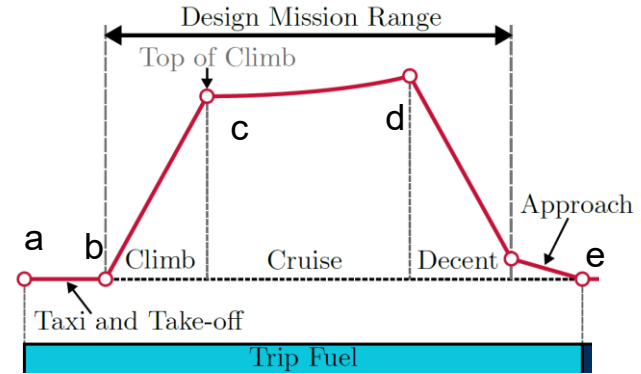
Velocity $\rightarrow$

Range $\rightarrow$

Specific fuel Consumption $\rightarrow$

Lift-to-drag Ratio $\rightarrow$

$$R = \frac{V}{C} \frac{L}{D} \ln \left(\frac{M_{i-1}}{M_i} \right)$$
$$\frac{M_i}{M_{i-1}} = \exp \left(\frac{-RC}{V \frac{L}{D}} \right)$$



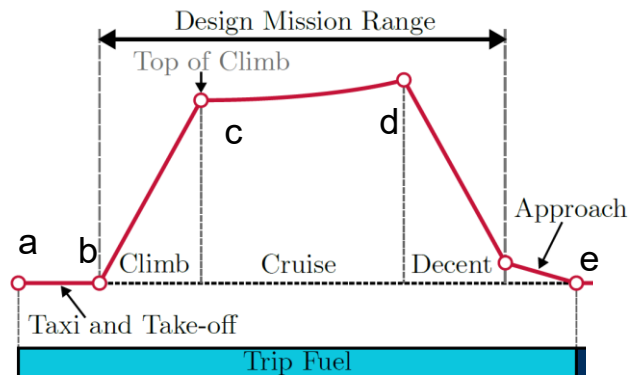
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- SFC** first guess – current engines

[CFM International LEAP - Wikipedia](#)

- L/D** first guess

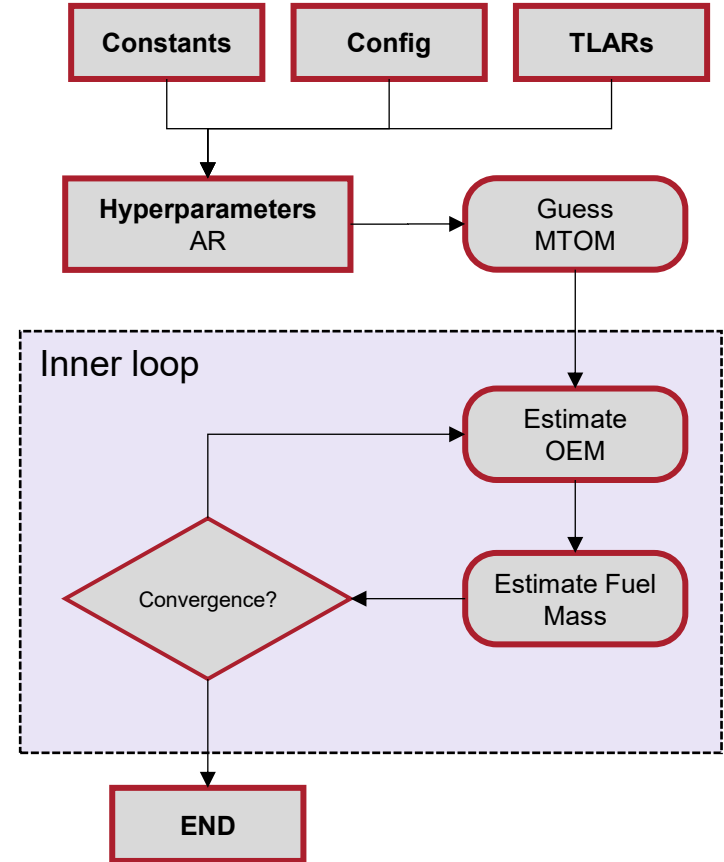
- Hyperparameter?
- Simple equation

$$15.5 \sqrt{\frac{A S_{ref}}{S_{wet}}} \text{ (Raymer 3.12)}$$

- Interaction with aerodynamics discipline!

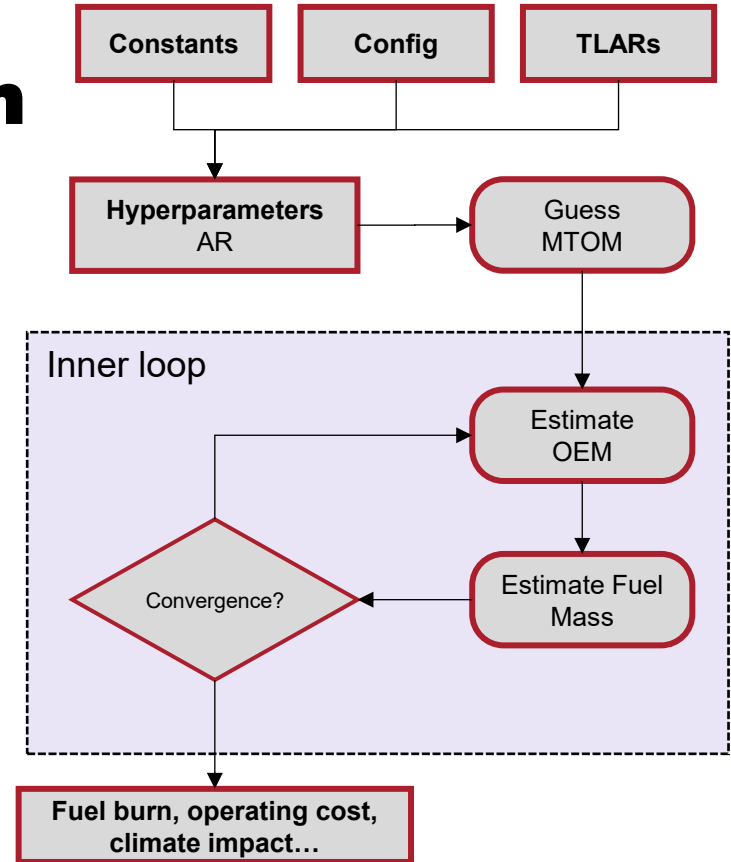
Putting it All Together

- To start with, we must first guess the MTOM! Many ways to do this ...
 - Historical averages
 - Fraction of payload, e.g. for B77F
$$\frac{M_0}{M_p} = 3.37$$
- You then iterate between estimating OEM and fuel mass until convergence
 - This will be a repeating theme!



Sizing Versus Optimisation

- For a given set of hyperparameters, you will get one output
 - This is **Sizing**
- By repeating the process and varying the hyperparameters, you will get different values of fuel burn, etc...
 - Increasing aspect ratio
 - Changing number of aircraft in fleet...
- Finding the set of hyperparameters that minimise your objective is **optimisation**
- During optimisation you will be running sizing multiple times...
- This will be very inaccurate with this fidelity of model....



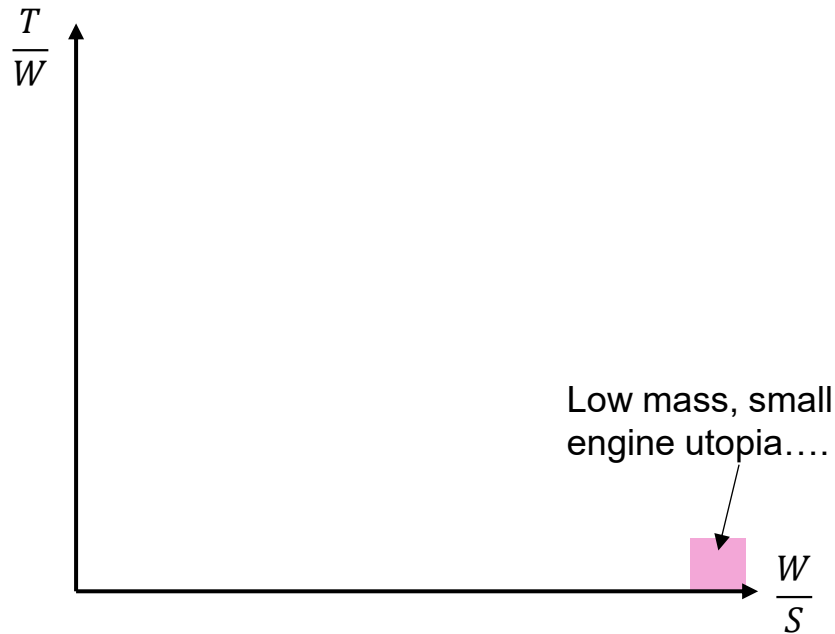
Increasing The Fidelity

Class ~II Methods

- At low-fidelity, everything is a fraction – at higher fidelities, everything becomes about geometry and physics.
 - “*Bottom-up Analysis*”
- Still aim to keep dimensionality low (TLARs + hyperparameters)
- We still want the problem to be deterministic!

Constraint Analysis

- Two important parameters
 - Thrust to weight ratio
 - Wing Loading
- These will define wing area and thrust (engine size) for a given MTOM
- Values are highly correlated
 - Big wing, low take-off speed, low thrust
 - Small wing, high take-off speed, large engines!
- Want to remove infeasible areas of the design space
 - Need equations like $\frac{T}{W} = f\left(\frac{W}{S}\right)$



Landing Constraint

- Landing wing loading limited
 - Independent of drag
- CS-25 stall speed (V_S) equals

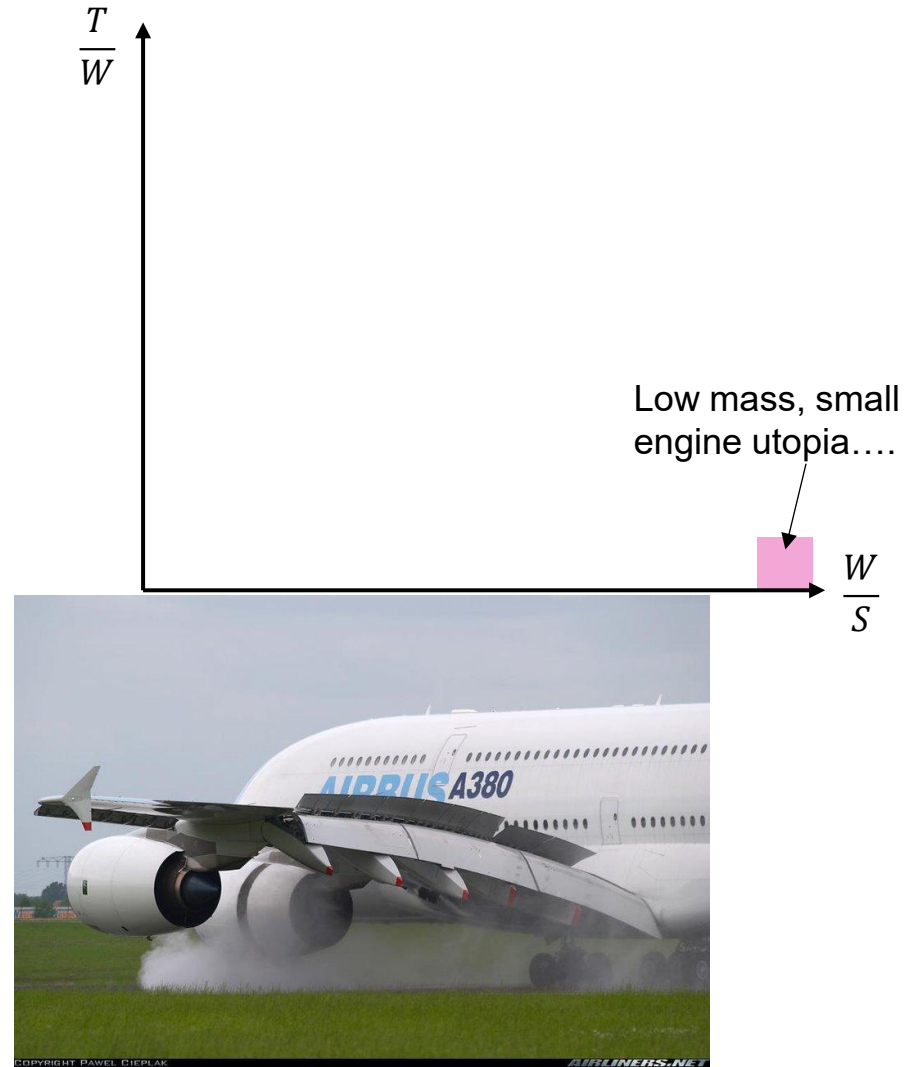
$$V_S = 1.3 V_{app}$$

- Where V_{app} is the approach speed
 - Maximum value set in your spec!
- From the classic lift equation

$$W = \frac{1}{2} \rho V_S^2 S C_{Lmax}$$

$$\frac{W}{S} = \frac{1}{2} \rho V_S^2 C_{Lmax}$$

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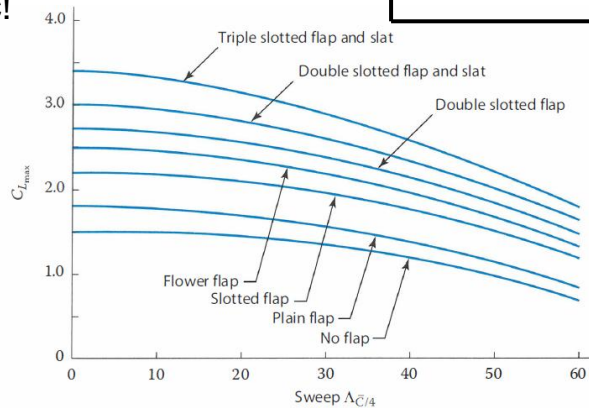
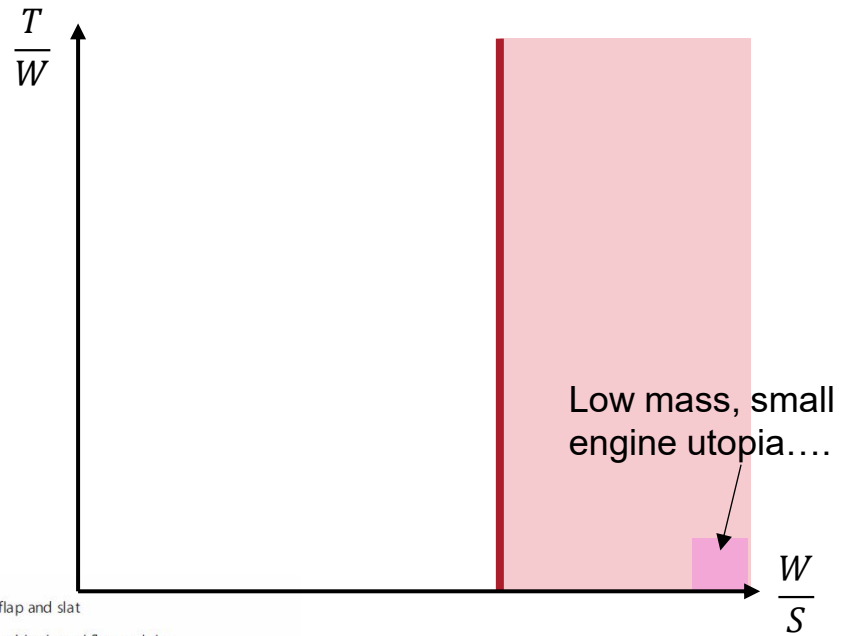


Fig. 5.3 Maximum lift coefficient.



- Estimating C_{Lmax} is tricky
 - Good luck aero discipline!
- To start with, you can use empirical relations
- Or make it a hyperparameter?

Take-off Constraint

- Take-off is perhaps the most complex dynamic event and aircraft goes through...
- For simplicity in early design, we can use a simplified model for ground roll distance

$$\frac{T}{W} = \frac{1.21}{g\rho C_{Lmax}^{TO} S_G} \left(\frac{W}{S} \right) + \frac{1}{2} \frac{C_D^{TO}}{C_{Lmax}^{TO}} + \frac{1}{2} \mu$$

Ground run

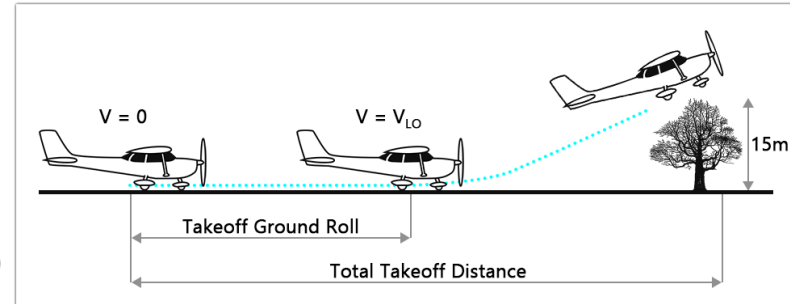
Ground friction constant (typ. 0.04)

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$\frac{T}{W}$



$\frac{W}{S}$



Take-off Constraint

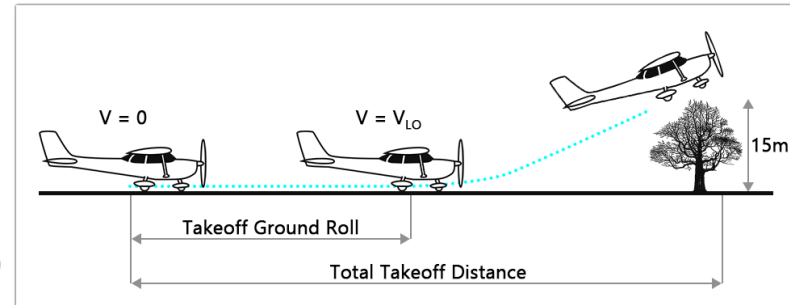
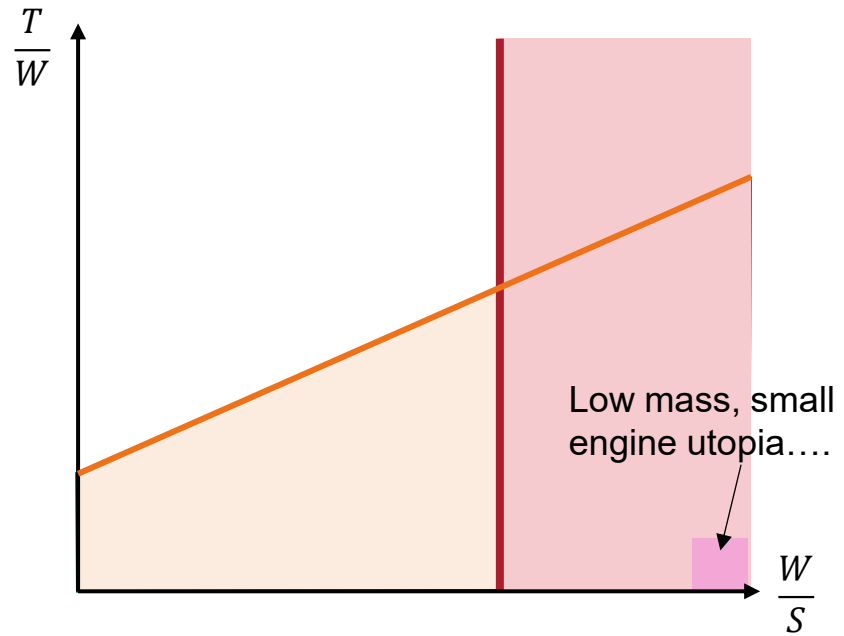
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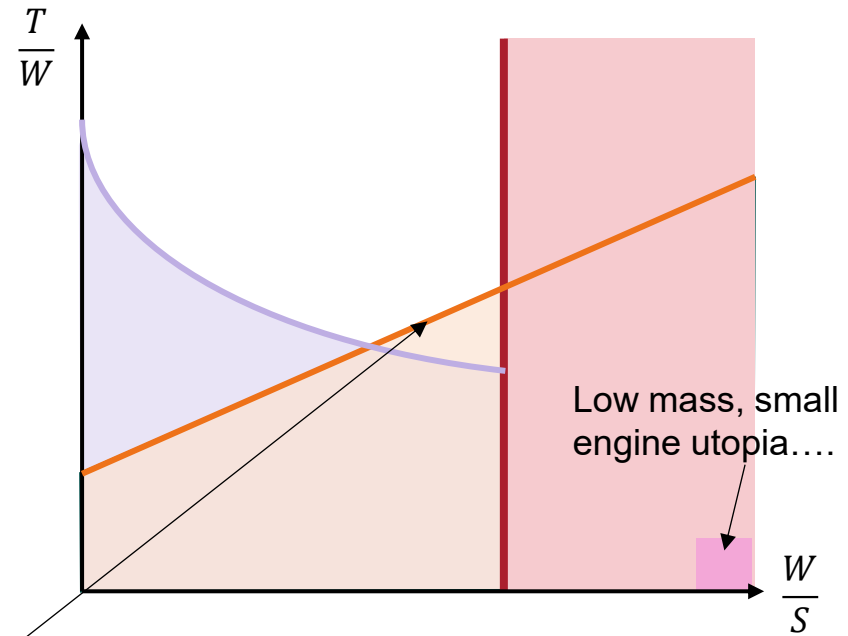
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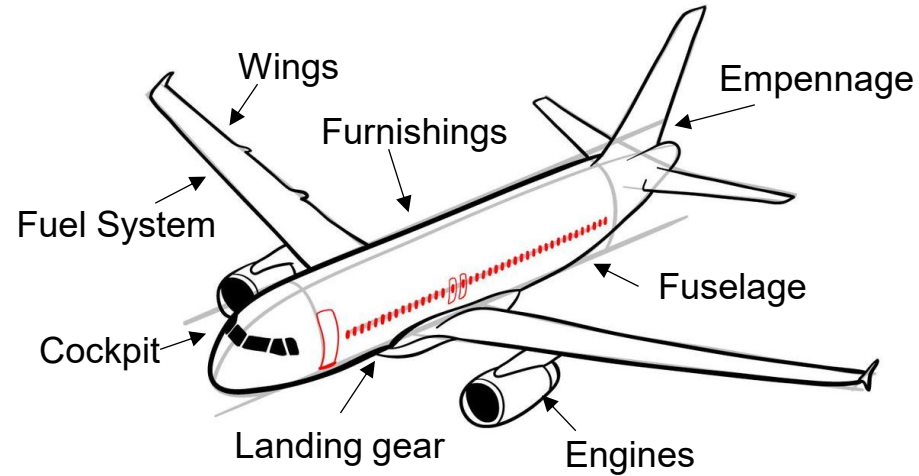
Other Constraints

- Similar Constraints can be constructed for:
 - Climb
 - Service ceiling
 - Coordinated turns
 - Engine out performance
 - (see Gundmunsson & Raymer...)
- Optimal designs live somewhere in the bottom left



Geometry Build Up

- Now that we have T/W and W/S , for a given MTOM, we can start to estimate the mass of basic components
- Summing these components up, we can get an estimate for the OEM
- To start with, I'll show you some statistical relations.
 - But one of the roles of each of the disciplines will be updating and improving these mass estimates!

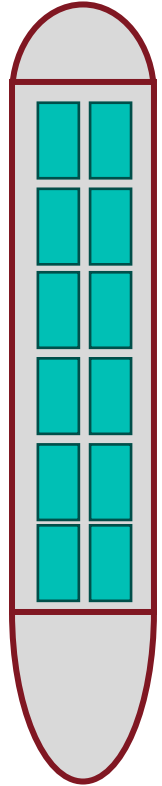


Fuselage

- For a given:
 - Payload, number of decks, fuselage width etc...
- You can estimate fuselage length
 - Cockpit fixed length, tail section ratio of diameter (see Raymer + example code)...
- You can then use empirical equations to estimate the mass

$$W_{\text{fuselage}} = 0.3280 K_{\text{door}} K_{\text{Lg}} (W_{\text{dg}} N_z)^{0.5} L^{0.25} S_f^{0.302} \\ \times (1 + K_{\text{ws}})^{0.04} (L/D)^{0.10} \quad (15.28)$$

Raymer



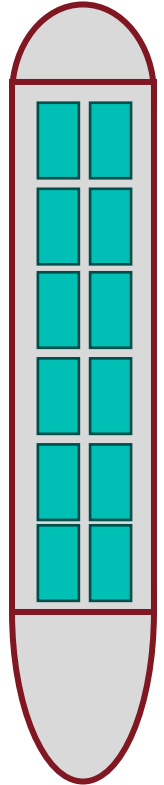
Fuselage

$$W_{\text{fuel system}} = 2.405 V_t^{0.606} (1 + V_i/V_t)^{-1.0} (1 + V_p/V_t) N_t^{0.5} \quad (15.34)$$

$$W_{\text{flight controls}} = 145.9 N_f^{0.554} (1 + N_m/N_f)^{-1.0} \\ \times S_{\text{CS}}^{0.20} (I_{\text{yaw}} \times 10^{-6})^{0.07} \quad (15.35)$$

$$W_{\text{APU installed}} = 2.2 W_{\text{APU uninstalled}} \quad (15.36)$$

$$W_{\text{instruments}} = 4.509 K_r K_{\text{tp}} N_c^{0.541} N_{\text{en}} (L_f + B_w)^{0.5} \quad (15.37)$$

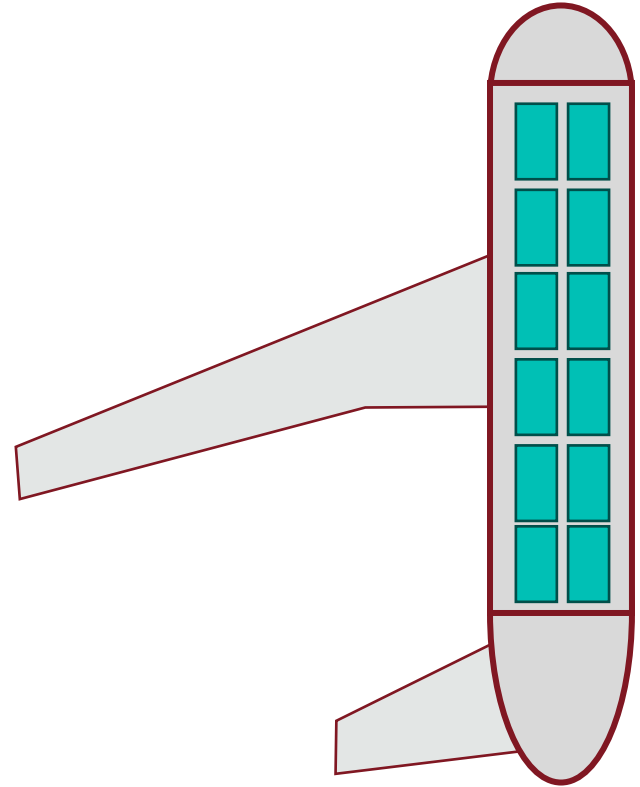


Lifting Surfaces

$$W_{\text{wing}} = 0.0051(W_{\text{dg}}N_z)^{0.557}S_w^{0.649}A^{0.5}(t/c)_{\text{root}}^{-0.4}(1+\lambda)^{0.1} \\ \times (\cos \Lambda)^{-1.0}S_{\text{csw}}^{0.1} \quad (15.25)$$

$$W_{\text{horizontal tail}} = 0.0379K_{\text{uht}}(1+F_w/B_h)^{-0.25}W_{\text{dg}}^{0.639}N_z^{0.10} \\ \times S_{\text{ht}}^{0.75}L_t^{-1.0}K_y^{0.704}(\cos \Lambda_{\text{ht}})^{-1.0} \\ \times A_h^{0.166}(1+S_e/S_{\text{ht}})^{0.1} \quad (15.26)$$

$$W_{\text{vertical tail}} = 0.0026(1+H_t/H_v)^{0.225}W_{\text{dg}}^{0.556}N_z^{0.536}L_t^{-0.5} \\ \times S_{\text{vt}}^{0.5}K_z^{0.875}(\cos \Lambda_{\text{vt}})^{-1}A_v^{0.35}(t/c)_{\text{root}}^{-0.5} \quad (15.27)$$



Etc.....

$$W_{\text{main landing gear}} = 0.0106 K_{\text{mp}} W_l^{0.888} N_l^{0.25} L_m^{0.4} N_{\text{mw}}^{0.321} N_{\text{mss}}^{-0.5} V_{\text{stall}}^{0.1} \quad (15.29)$$

Landing gear

$$W_{\text{nose landing gear}} = 0.032 K_{\text{np}} W_l^{0.646} N_l^{0.2} L_n^{0.5} N_{\text{nw}}^{0.45} \quad (15.30)$$

Extras

$$W_{\text{hydraulics}} = 0.2673 N_f (L_f + B_w)^{0.937} \quad (15.38)$$

$$W_{\text{electrical}} = 7.291 R_{\text{kva}}^{0.782} L_a^{0.346} N_{\text{gen}}^{0.10} \quad (15.39)$$

$$W_{\text{avionics}} = 1.73 W_{\text{uav}}^{0.983} \quad (15.40)$$

$$W_{\text{furnishings}} = 0.0577 N_c^{0.1} W_c^{0.393} S_f^{0.75} \quad (15.41)$$

(does not include cargo handling gear or seats)

$$W_{\text{air conditioning}} = 62.36 N_p^{0.25} (V_{\text{pr}}/1000)^{0.604} W_{\text{uav}}^{0.10} \quad (15.42)$$

$$W_{\text{anti-ice}} = 0.002 W_{\text{dg}} \quad (15.43)$$

$$W_{\text{handling gear}} = 3.0 \times 10^{-4} W_{\text{dg}} \quad (15.44)$$

$$W_{\text{military cargo handling system}} = 2.4 \times (\text{cargo floor area, ft}^2) \quad (15.45)$$

Engines

- As an initial guess, we can use **Rubberised Engines**
- Start from a known engine, then scale thrust, dimensions and mass to suit

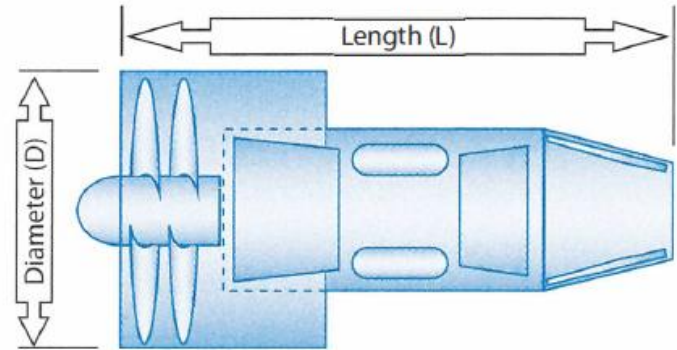
$$T = T_{act}(SF)$$

$$L = L_{act}(SF)^{0.4}$$

$$D = D_{act}(SF)^{0.5}$$

$$M = M_{act}(SF)^{0.4}$$

Raymer
(section 10)



Improved Mission Analysis

- To improve the accuracy of the mission analysis, we can
 - incorporate mass fraction calculation for more segments
 - Include an alternate mission to estimate the required reserve fuel
 - Improve aero estimates
 - With help from the aero discipline!

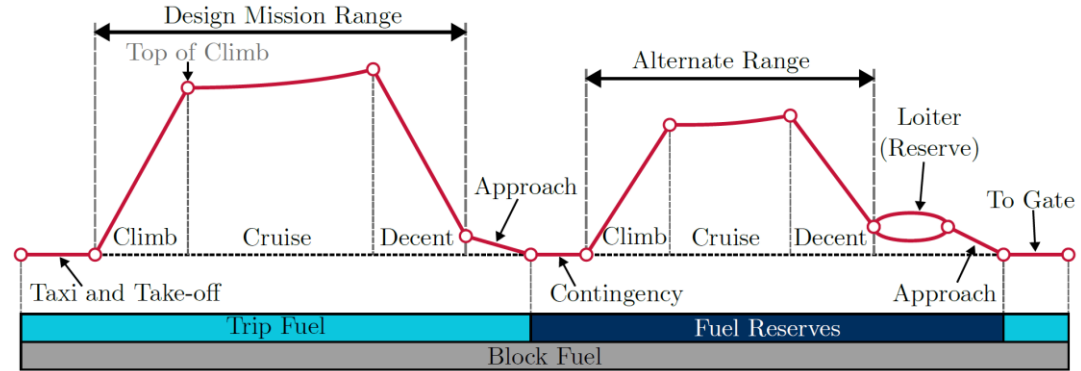
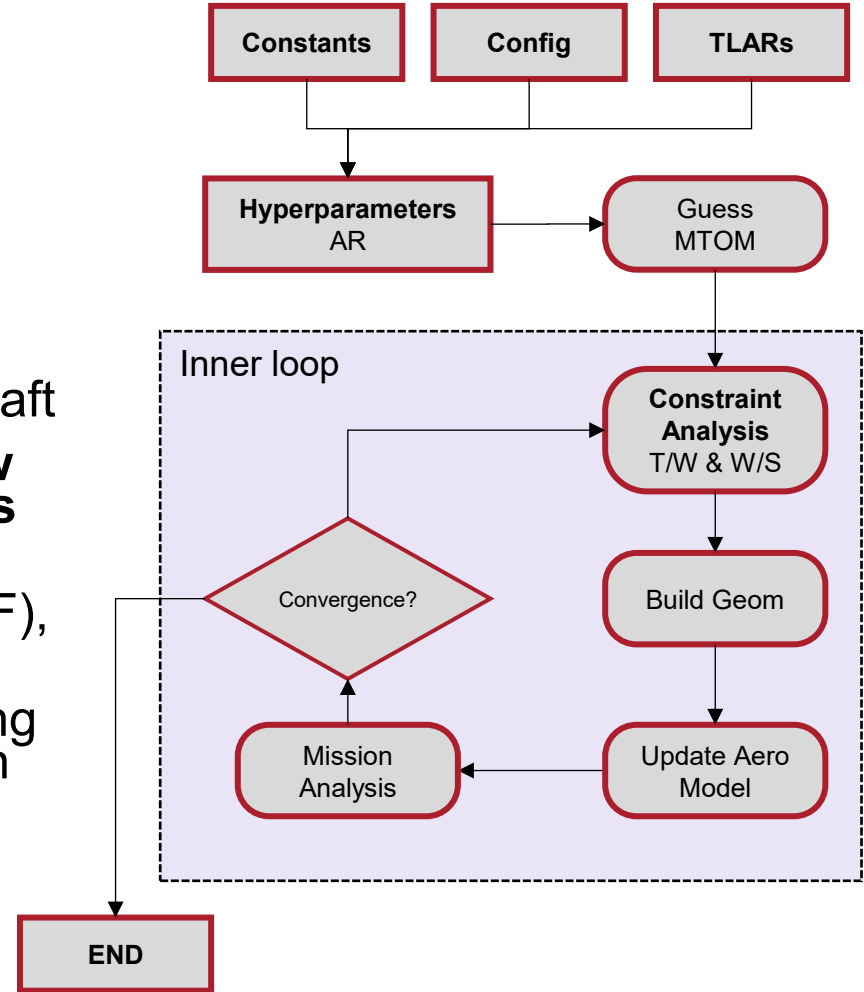


Figure 4: Design mission segment and fuel breakdown

Putting it All Together

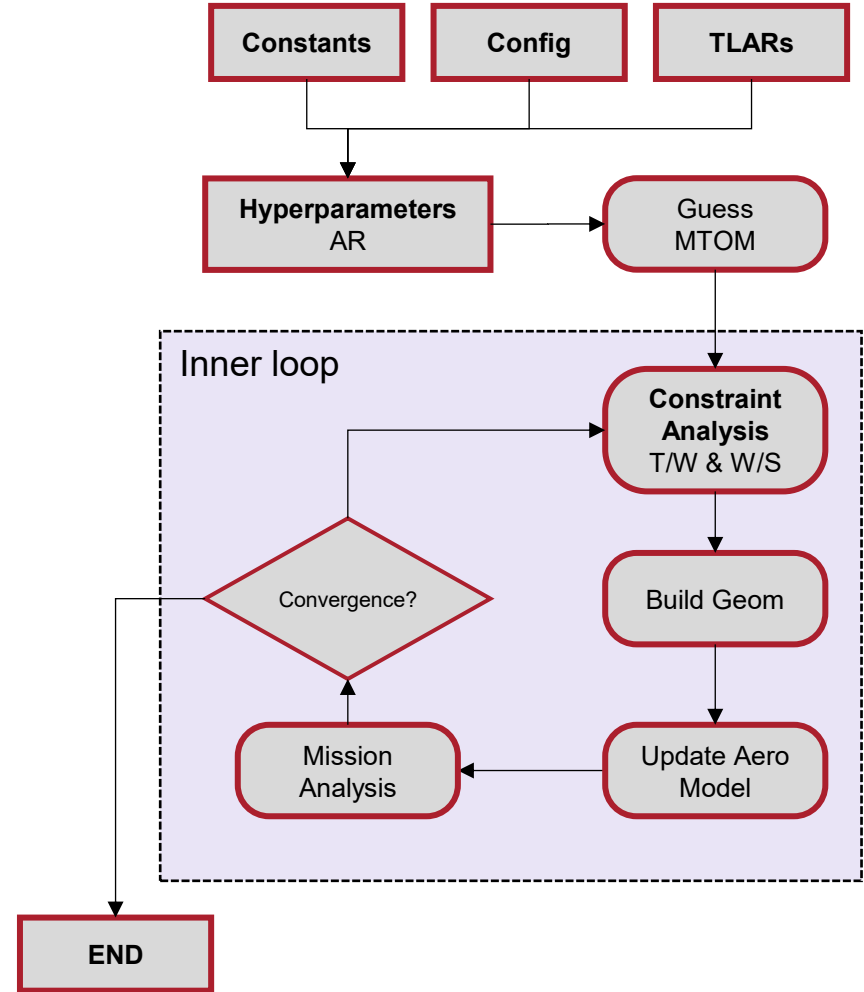
- Same process as before
 - Guess MTOM and iterate until convergence!
- Remember you're not building an aircraft
- **You're building a tool to assess how the performance of an aircraft varies as you change parameters!**
- Anchor your tool to known data (B777F), then start varying parameters
- Be sceptical, there will be lots of moving parts. Try to build confidence that each part is functioning properly.



Help is at Hand!

- I've done a lot of the initial work for you!
 - See the Sizing tool linked on Blackboard!
- Have a look through this tool and identify your discipline within it
- We'll go through this tool in detail next week!

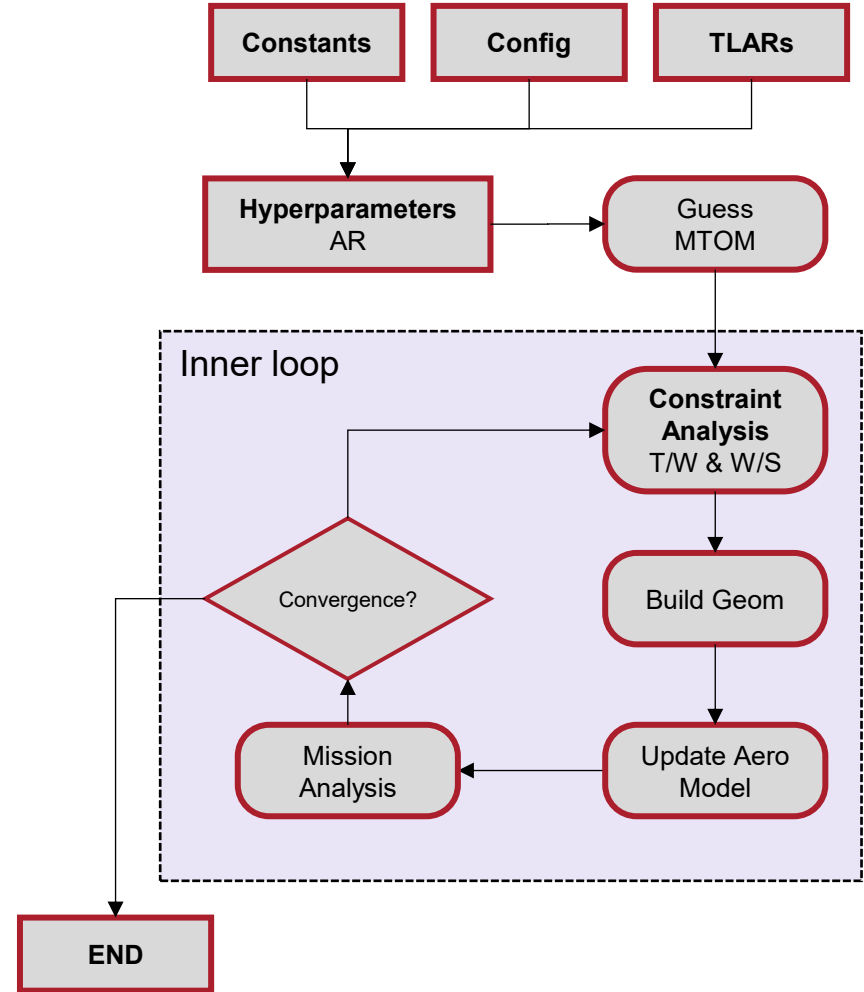
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Conclusions

Conclusions

- Explored how to size a “clean sheet” design to estimate mission performance
- Explored estimating T/W and W/S using constraint diagrams
- Explored Mission Analysis using the mass fraction method



Current Tasks

- Meet your TSRs
- Work in your teams towards completing the Concept Down-selection report

Questions?